**Department of Electrical, Computer, & Biomedical Engineering**

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| Course Number | ELE734 |
| Course Title | Low Power Digital Integrated Circuits |
| Semester/Year | F2025 |
| Instructor | Andy Ye |
| TA Name | Esraa Alaa Aldeen |

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|---|---|
| Lab/Tutorial Report No. | 1 |

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|---|---|
| Report Title | Characteristic Of MOSFET Devices |

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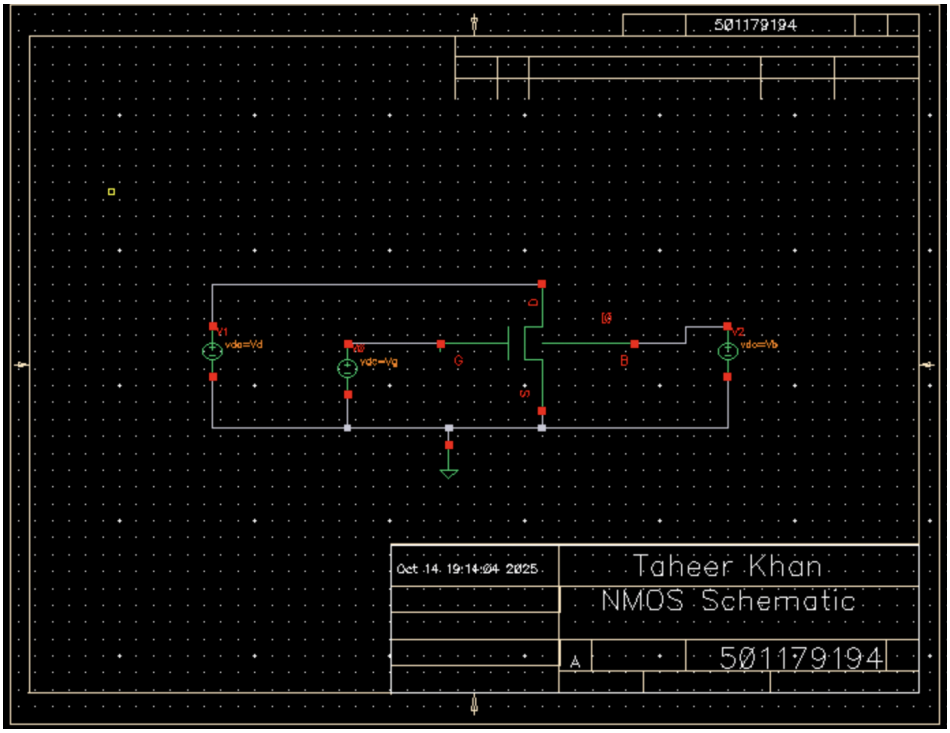
|---|---|
| Section No. | 5 |
| Submission Date | October 17, 2025 |
| Due Date | October 17, 2025 |

| Student Name | Student ID | Signature* |
|---|---|---|
| Taheer Khan | 501179194 | TK |

*By signing above you attest that you have contributed to this written lab report and confirm that all work you have contributed to this lab report is your own work. Any suspicion of copying or plagiarism in this work will result in an investigation of Academic Misconduct and may result in a “0” on the work, an “F” in the course, or possibly more severe penalties, as well as a Disciplinary Notice on your academic record under the Student Code of Academic Conduct, which can be found online at: www.ryerson.ca/senate/current/pol60.pdf.

## **1.1** Schematic of NMOS & PMOS Transistors:



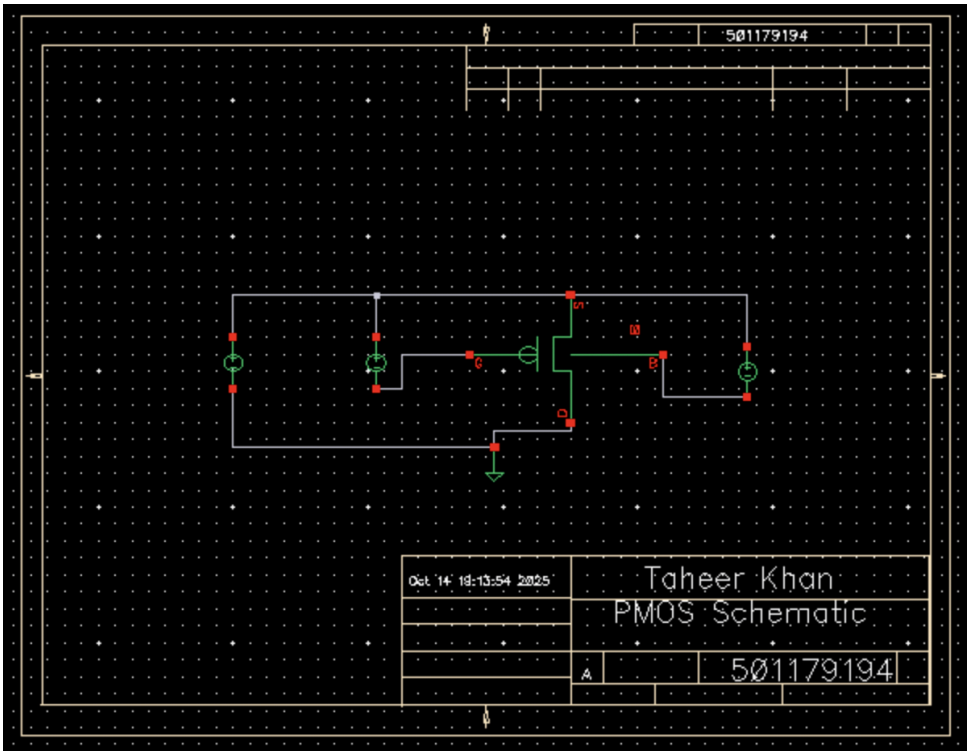




**Table 1:** Schematic for NMOS & PMOS Transistors

## **1.2** Family of I-V Characteristic Waveforms for NMOS and PMOS Transistors:

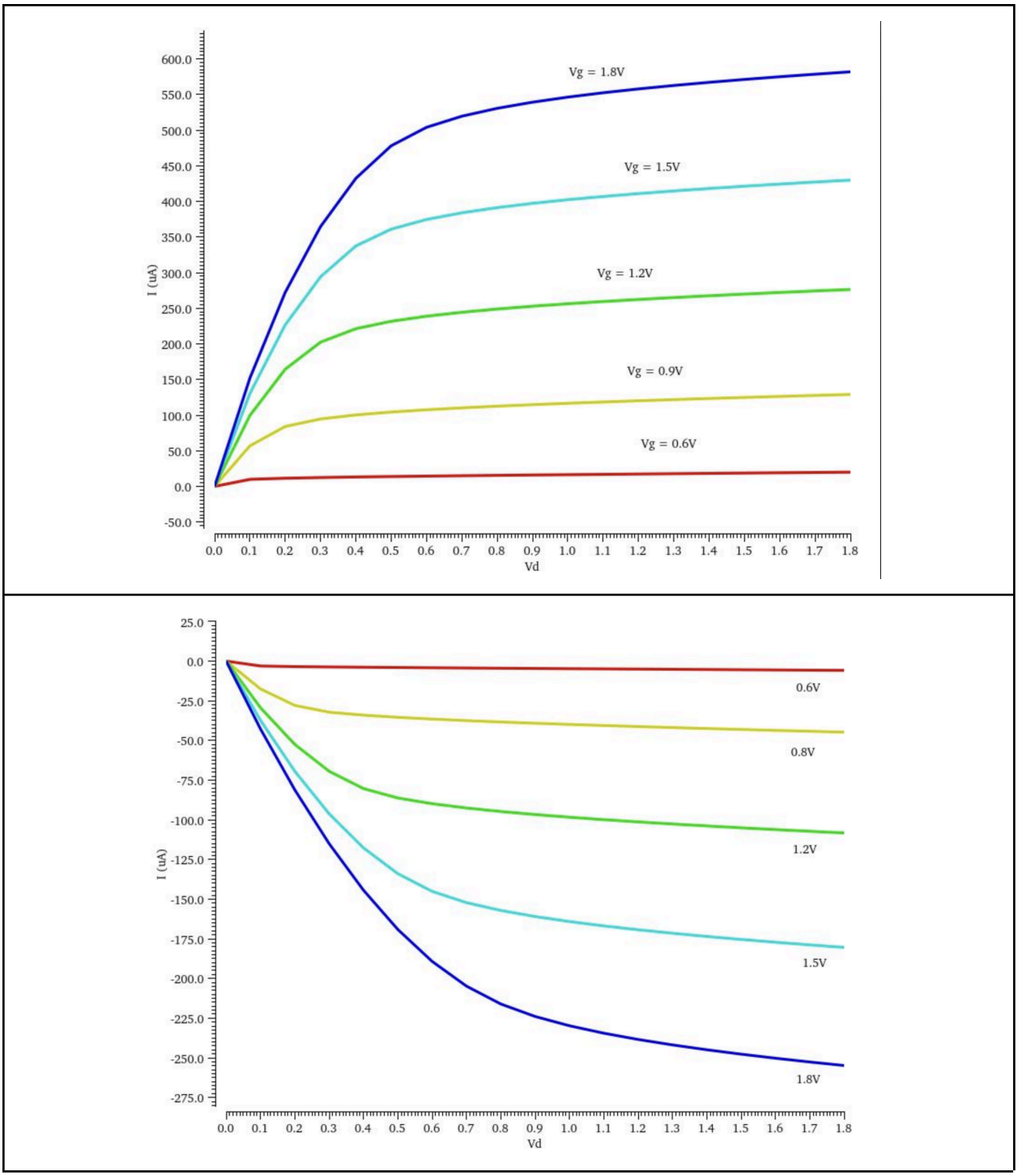



**Table 2:** I-V Characteristic waveforms for NMOS & PMOS (Respectively).

## **1.3** Subthreshold Conduction for NMOS & PMOS Transistors:

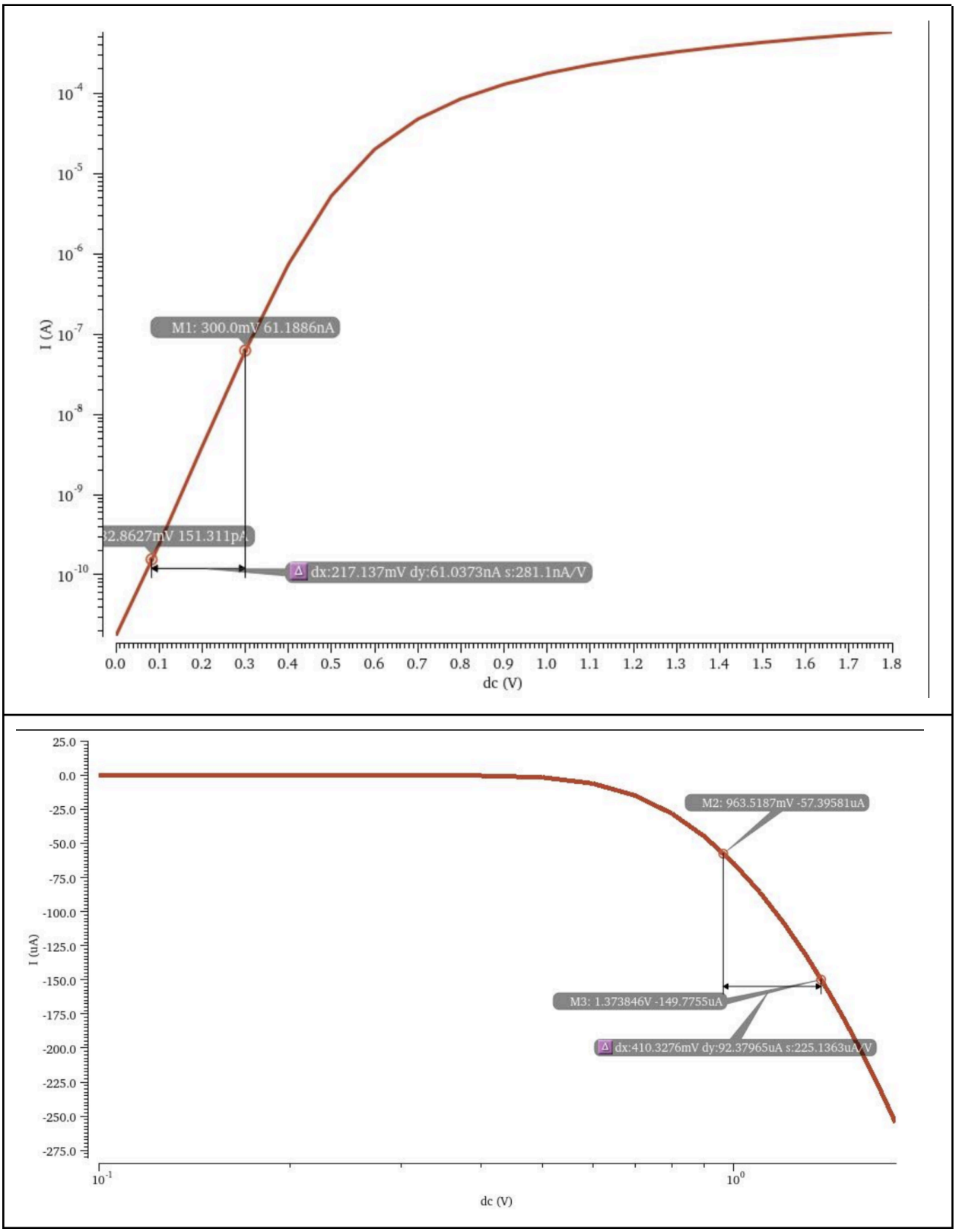
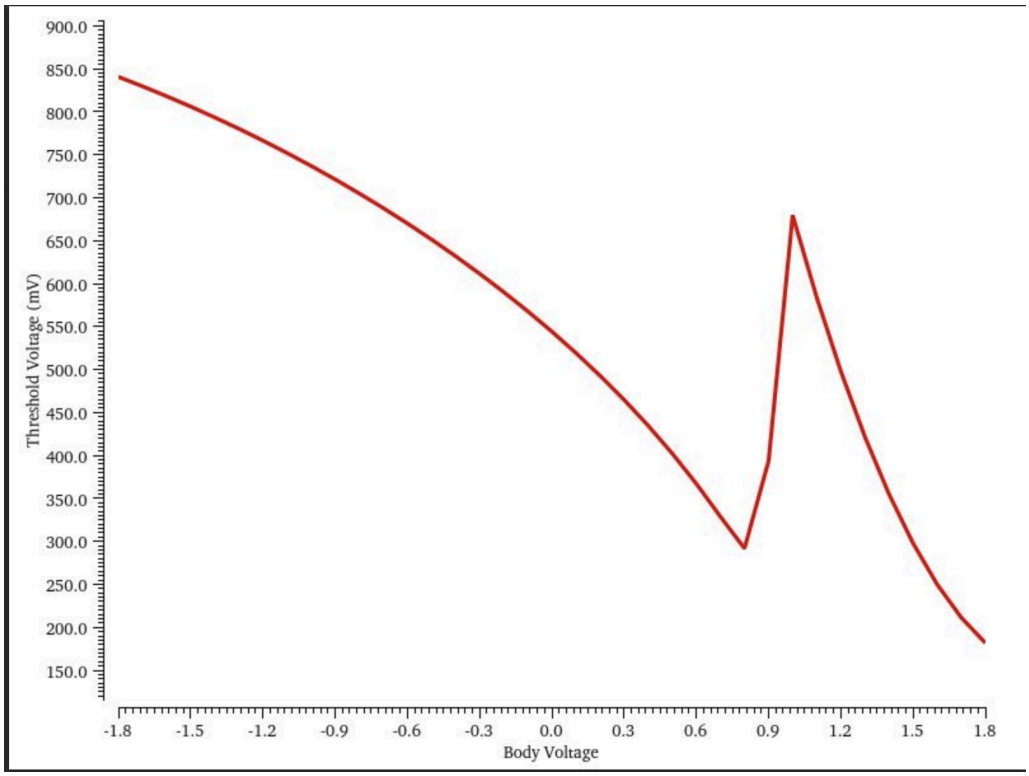



**Table 3:** Subthreshold Conduction for NMOS & PMOS (Respectively)

## **1.4** Body Effect Analysis of NMOS & PMOS Transistors:



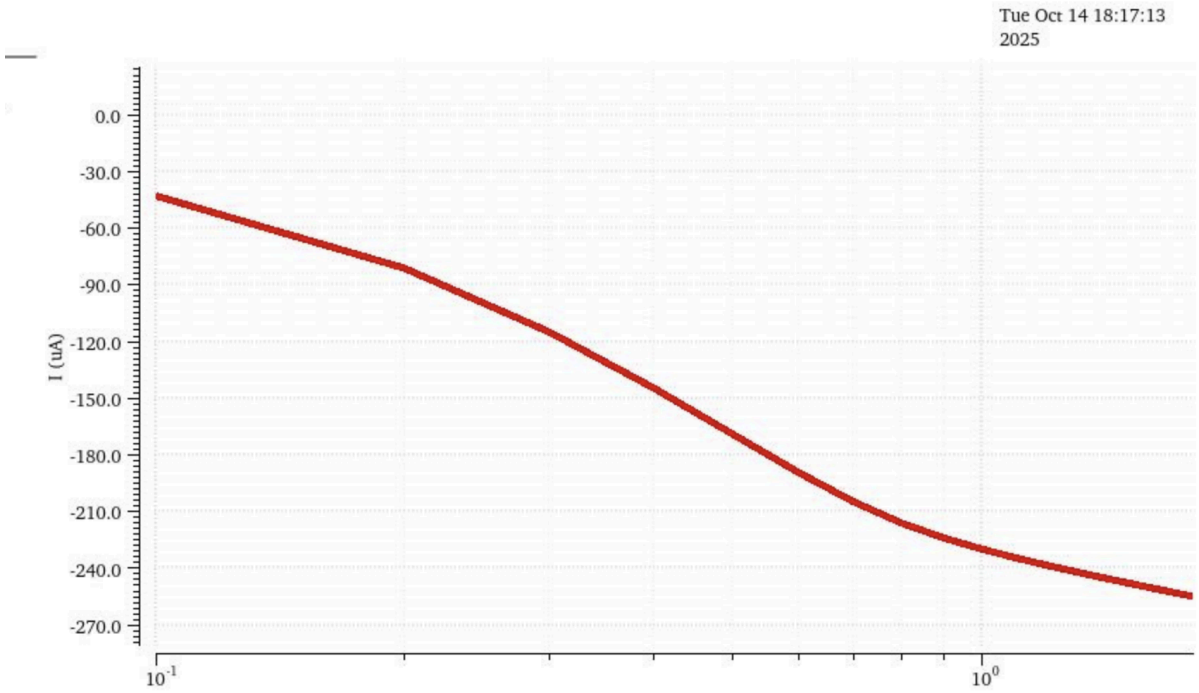
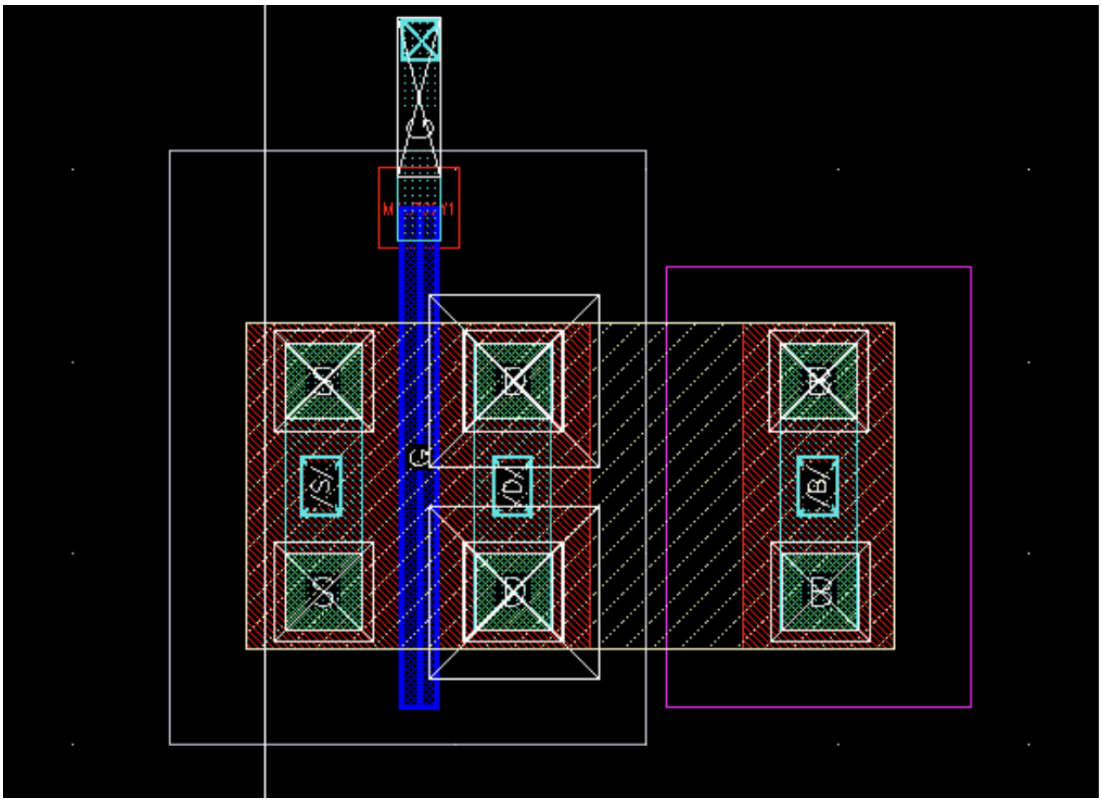




**Table 4:** Body Effect Analysis for NMOS & PMOS (Respectively)

## **1.5** Layout & Extracted View of NMOS & PMOS Transistors:



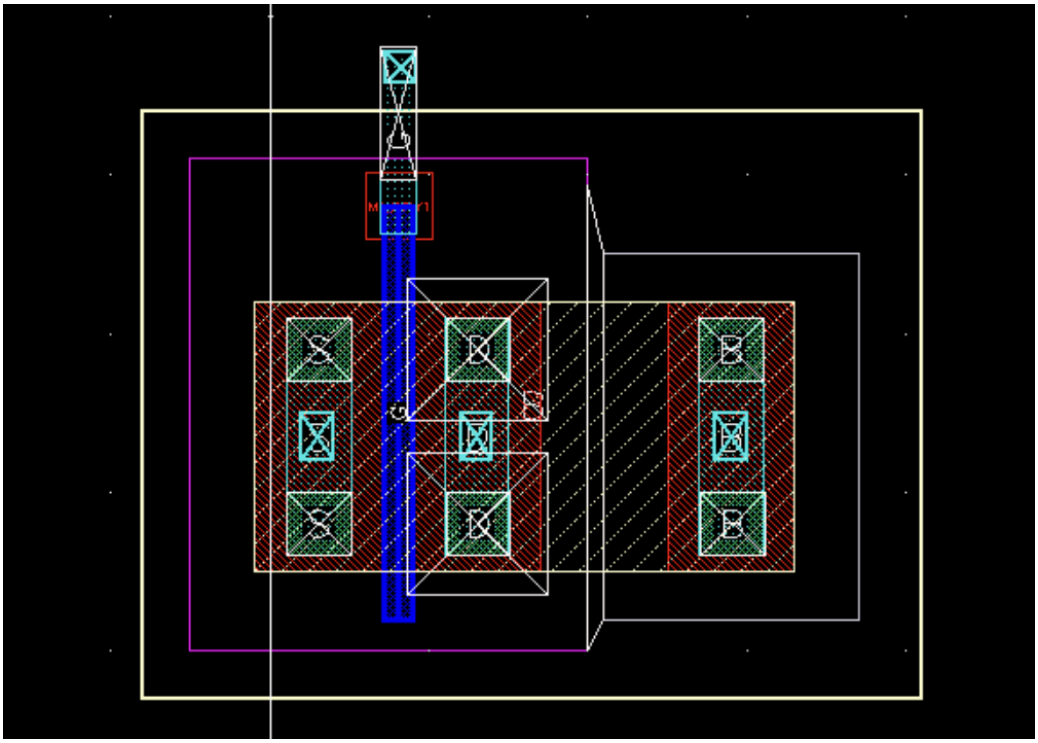



**Table 5:** Layout for NMOS & PMOS (Respectively)

**1.6** I-V Characteristics of Waveforms for Schematic & Extracted view for NMOS & PMOS Transistors:

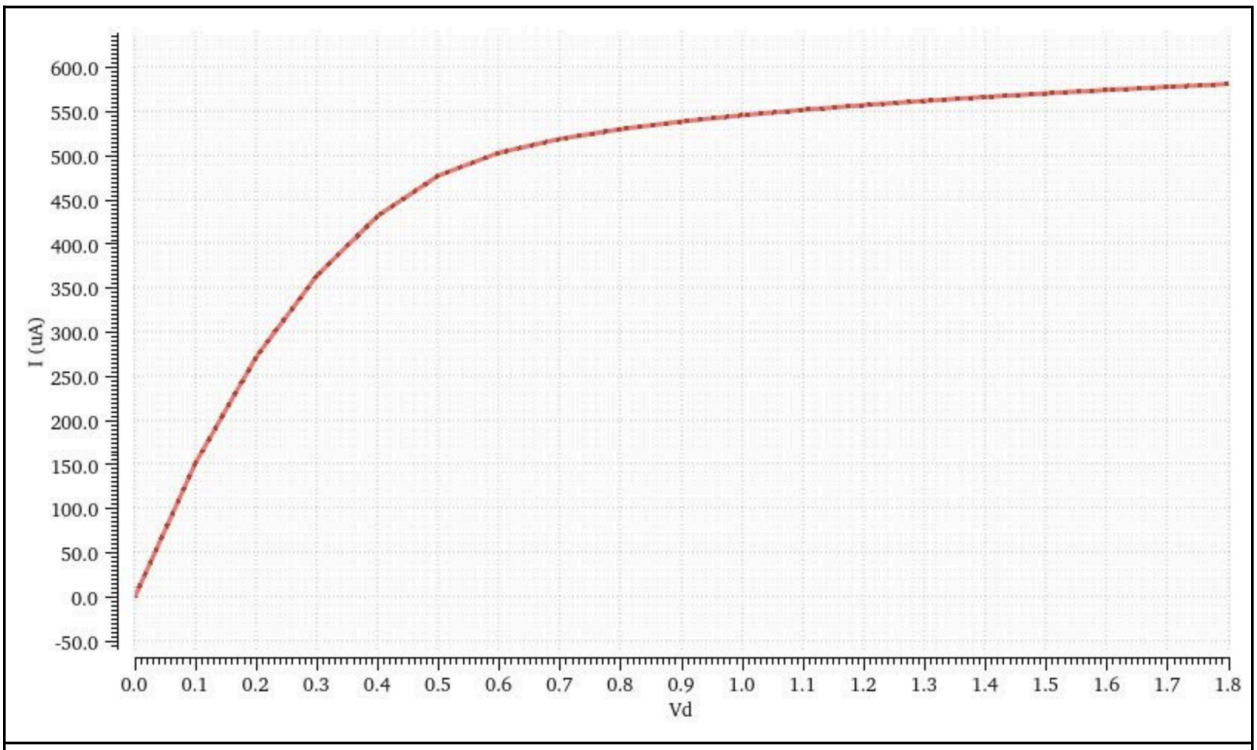



**Table 6:** I-V Characteristic of Schematic & Extracted View for NMOS & PMOS (Respectively)

## **1.7** Conclusion:

The family of I–V characteristic curves for both NMOS and PMOS transistors displayed the expected behavior: as the gate to source voltage increased, the drain current Id rose sharply before saturating, marking the transition from the linear to saturation region. The extracted device simulations closely followed the schematic results, with minor deviations attributed to parasitic resistances and capacitances introduced during layout extraction. These parasitics slightly reduced the drain current in the extracted simulations, confirming the practical impact of non-ideal device effects.

The subthreshold and body-effect analyses showed mixed results. The subthreshold slope confirmed the expected exponential current decay when the gate-to-source voltage was below the threshold voltage. However, the Vth–VSource to Body curve did not follow the expected trend, suggesting the body effect did not behave as predicted. This discrepancy may have been caused by setup issues in Cadence, such as incorrect body terminal connections or overlooked model parameters. Deviations from prelab values were mainly due to these potential setup errors, rounding inaccuracies, and parasitic effects in the extracted transistor.

Overall, the results demonstrated strong agreement between theory and simulation, verifying that both the schematic and layout implementations were correctly designed and that second-order effects realistically influenced transistor performance.